

STATEMENT OF WORK

Business Requirements Document (BRD)

Lesson Learned Form

Quality Module – Web & Mobile Application

Abstract

This SOW defines the scope, functional requirements, workflow, roles and deliverables for the Lesson Learned Form. The system covers the complete process of Lesson Learned form - through web and mobile platforms. It aims to improve traceability, trigger action workflows, and create a centralized learning database across projects.

Document Title	SOW – Business Requirements Document Lesson Learned Form
Document Reference	Lesson Learned Form-SOW-V2
Version	v2.0
Prepared By	Subbu Sankar V R
Reviewed By	
Approved By	
Date	02-07-2026
Sensitivity	Internal Use Only
Status	For Review

1. Document Control

1.1 Revision History

Version	Date	Author	Reviewed By	Description of Changes
v1.0	01-07-2026	Subbu Sankar V R	Mohammed Attaullah	The SOW has been fully updated to address all identified gaps. The revised document now includes user controls for AI-suggested text, offline mobile syncing capabilities, and expanded limits of up to five images for uploading. Also revised the L1/L2 escalation hierarchy, restricted signatures to secure system auto-captures, and embedded the missing flowchart.
v2.0	02-07-2026	Subbu Sankar V R		

2. Introduction

Algihaz shall record lessons learned at the projects which can be both positive and negative. These lessons improvise the projects in terms of their quality compliance and executional methodology.

- Enable the project to be proactively ready for the possible setbacks.
- Understand the bottlenecks which hamper the execution process.
- The root Cause analysis and corrective action can be used across other similar project sites, which makes this centralised database as knowledge hub.

The proposed solution digitizes the Lesson Learned Form creation process and eases the trackability of the created versions. This Statement of Work (SOW) outlines the scope, features, responsibilities, and deliverables for the design, development, and deployment of the Lesson Learned Form.

3. Objectives

The Lesson Learned Form application shall achieve the following objectives:

- Record all the relevant information of lesson learned form and trigger the workflow for the approval without any complexity.
- Enable complete transparency and visibility across the project sites to create a form without anyone dependency.
- Encourage the stakeholders to capture more Lesson Learned forms with its simple and precise UI
- Reduce manual textual inputs wherever possible by providing dropdowns and radio buttons.
- Timely notifications to the concerned stakeholders as soon as a process is completed.
- Create and submit a form instantly with all the real time inputs.

4. Executive Summary

4.1 Project Background

Currently the Lesson Learned Form creation follows a manual methodology where a user enters the relevant details in the standard report and the same is being sent to the approver for review.

As the present methodology does not have a trackable and actionable platform to search the created forms, a digital solution for the Lesson Learned Form is proposed which captures all the corresponding details and makes it reusable across other projects.

4.2 Key Pain Points & Gap in Current Process

#	Pain Point / Gap	Impact
1	Reference Number	A same reference number can be used for two forms making the data management part very tedious.
2	Basic Details	Date and Location can be forged with fake values which can cause serious compliance issues.
3	Core Details	The form creator needs to manually type all the details causing human error in the description and corrective actions.
4	Photo Upload	The photos should have been taken prior to the form creation and can lead to the submission of wrong image also.
5	Submission & Approval	Form creator and approver should provide their details and signature manually. Also, the details are communicated through email.
6	Send back Workflow	If the provided details are not sufficient, form approver again sends email asking for proper response which makes tracking and data management very difficult.
7	Centralised Database	The lesson learned forms are nowhere accessible by the other stakeholders hampering the knowledge sharing process.

5. Workflow Overview

The Lesson learnt form creation consists of following stages.

STAGE 1	STAGE 2	STAGE 3
Form Creation	Form Submission	Form Approval

5.1 Stage 1

Stage 1 – Form Creation

- The Form Creator opens a New Lesson Learned Form

- Reference number will be autogenerated, and the other basic details will be fetched from the system.
- Enters the core details with minimal basic details and uploads the photographs.

Stage 2 – Form Submission

- The system auto captures the form creator details from the master and updates his details including signature.
- Post form submission, system triggers notification to the form approver in email.

Stage 3 – Form Approval

- The form approver opens the received form, checks the basic details and accords their approval.
- If the provided details are not sufficient, approver can send back the form to the creator asking for more details.
- Workflow will continue further until the approver approves the form.

6. Application Modules

Feature	Description	Module
Lessons Learned Form Management	Create and manage all the Lessons Learned forms	Web & Mobile
Reference Number	Auto-generate the reference number based on the provided format.	Web & Mobile
Standardised Dropdowns	Discipline, Categorization & Impact — all from configurable masters.	Web & Mobile
Basic Details	Auto captures the date and location from the system points.	Web & Mobile
Core Details	Allow users to enter responses for the Description, Root Cause, Correction, and Corrective Action fields. Based on the user's input, the AI should automatically rephrase the content into clear, grammatically correct, and professionally written sentences. The form creator must have the option to Accept, Edit, or Reject the AI suggestion before final submission.	Web & Mobile
Before/After Photographs	Evidence attached directly from device camera or storage.	Web & Mobile
Submission & Approval	Auto capture Designation, Name, Signature from user masters.	Web & Mobile
Role-Based Access Control	Defined roles with clear permission segregation.	Web & Mobile
Reports & Analytics	Lesson Learned Report as per the provided format, Total Lessons Learned, Positive vs Negative Trends, Lessons learned summary	Web & Mobile
Data Export	PDF/Excel export	Web & Mobile
Push Notifications	Real-time alerts for form creators and approvers after completion of every process.	Web & Mobile

7. Application Fields

The following fields will be captured and all fields marked with * are mandatory.

Stage 1 – Form Creation

Field	Type	Source	Mandatory
Project Name	Dropdown	List of Projects assigned to user	Yes
Reference Number	Auto generated	Format has been requested from Document control team.	Yes
Title	Text Field	Manual	Yes
Discipline	Dropdown	Master	Yes
Categorization	Dropdown	Master	Yes
Date	Auto generated	Current Date	Yes
Location	Auto generated	GPS Coordinates	Yes
Issue Category	Dropdown	Minor, Moderate, Major	Yes
Impact	Radio Button	Positive, Negative	Yes
Description	Text field	Manual	Yes
Reference	Text Field	Manual	Optional
New / Repeated Issue	Radio	New, Repeated	Yes
No of times repeated	Numerical Field	Manual	Mandatory if issue is repeated
Where was it repeated?	Text Field	Manual	Mandatory if issue is repeated
Root Cause	Text Field	Manual	Yes
Correction	Text Field	Manual	Yes
Corrective Action	Text Field	Manual	Yes
Before Photo	File upload	From Device/Cam	Yes
After Photo	File upload	From Device/Cam	Yes
Remarks	Text Field	Manual	Optional

Stage 2 – Form Submission

Field	Type	Source	Mandatory
Designation	Auto generated	Master	Yes
Name	Auto generated	Master	Yes
Signature	Auto generated	Master	Yes
Date	Auto generated	Master	Yes

Stage 3 – Form Approval

Field	Type	Source	Mandatory
-------	------	--------	-----------

Approval	Radio Button	Yes, No	Yes
Remarks	Text	Manual	Mandatory if Approver rejects.
Designation	Auto generated	Master	Yes
Name	Auto generated	Master	Yes
Signature	Auto generated	Master	Yes
Date	Auto generated	Master	Yes

8. Escalation Matrix

Escalations are triggered automatically based on the issue category and impact involved in the lesson learned. Subsequent escalation levels are defined below if suitable corrective action has not been taken within the stipulated time.

Priority	Risk Combination	SLA(Days)	Escalation Action
P1	Major + Impact (Negative)	Triggered when remains unresolved for 2 days.	To: Approver CC: Form Creator If overdue → auto upgrade to L1

Level	Risk Combination
L1	Triggered when P1 remains unresolved for 3 days. Escalated to Project Manager keeping form creator and approver in CC
L2	Triggered when L1 remains unresolved for 3 days. Escalated to BU Head/Director for management intervention keeping Project Director, form creator and approver in CC every 2 days until the form is closed.

9. Role-Based Access Control (RBAC)

Role	Phase/Access	Permissions & Responsibilities
Form Creator	Form Creation Form Submission	- Create and manage Lessons Learned Report. - Submit the form to the approver - Revise the form if approver gives any remarks.
Form Approver	Form Approval	- Receives the Lesson Learned form from the Form Creator. - Approves/Rejects the created forms.
BU Head/ Project Head/Quality Department	All Phases	View all the created forms.
Admin	All Phases	Responsible for the complete governance of Lessons Learned Form application.

10. Configuration & Masters

The system shall be designed for easy adaptation to new projects & departments.

Master/Config Team	Type	Remarks
Business Unit	One time setup	Organisation's list of BU
Project Master	One time setup	- Name, - SBG - BU, - Project Group (List of Projects to be provided)
User Master	One time setup	- Name, - Role (RBAC), - Email (List of Users to be provided)
Discipline Master	Configurable	- Electrical - Civil - Mechanical - Precast - Architectural - Testing Commissioning - Plumbing - RTR & Piping - Instrumentation - Communications - Storage & Handling - Material Receiving
Categorization Master	Configurable	- Company - Third Party - Supplier
Escalation Matrix	Configurable	SLA days, upgrade days and roles.

11. Functional Requirements as User Stories

ID	Module	Requirement	Acceptance Criteria
FR-1	Reference Number	As a form creator, when I open the New Form option, I want the reference number to be auto generated so that uniformity is maintained across the organization.	The system must successfully generate the reference number as per the configured format. <i>(Once the DC team provides the reference number format, the same will be considered here)</i>
FR-2	Basic Details	As a form creator, I want to enter basic details such as Title, Date, and Location so that I can create a Lesson Learned record.	The system must capture the user inputs successfully. The application must save the entered details.
FR-3	Standardized Dropdowns	As a form creator, I want to choose the relevant option from the Discipline, Category, and Issue Category dropdowns so that I can	The system must display all available values for Discipline, Category, and Issue Category from the master data.

		select the correct option and maintain standardized values.	The application must record the selected values successfully.
FR-4	Core Details Impact	As a form creator, I want to select the type of impact using radio buttons so that the effect of the issue can be clearly identified.	- The system must allow the form creator to select either Positive or Negative. - Only one option can be selected at a time.
FR-5	Core Details Description	As a form creator, I want to enter the Description of the Lesson Learned so that readers can clearly understand the issue.	- The system must allow the form creator to enter the description. - The application must accept all valid textual input. - The system must rephrase the entered text into a clear, grammatically correct, and well-structured sentence using AI, even if only a short sentence is entered. - The form creator must have the option to Accept, Edit, or Reject the AI suggestion before final submission.
FR-6	Core Details References	As a form creator, I want to provide reference details, if any, so that supporting documents can be referred to for better clarity.	The system must allow the form creator to enter reference details. Reference details should be optional.
FR-7	Core Details Issue Repetition	As a form creator, I want to specify whether the issue has occurred previously so that the severity of the issue can be assessed.	- The system must provide Yes/No options to indicate whether the issue has been repeated. - If Yes is selected, the system must prompt the user to enter the number of times the issue is repeated and a text box to provide further details. - If No is selected, the additional field must remain hidden.
FR-8	Core Details Root Cause	As a form creator, I want to enter the Root Cause so that I can explain why the issue occurred.	- The system must allow the form creator to enter the root cause. - The system must rephrase the entered text into a clear, grammatically correct, and well-structured sentence using AI, even if only a short sentence is entered. - The form creator must have the option to Accept, Edit, or Reject the AI suggestion before final submission.
FR-9	Core Details Correction	As a form creator, I want to enter the Correction so that I can explain how the issue was rectified.	- The system must allow the form creator to enter the correction. - The system must rephrase the entered text into a clear, grammatically correct, and well-structured sentence using AI, even if only a short sentence is entered.

			The form creator must have the option to Accept, Edit, or Reject the AI suggestion before final submission.
FR-10	Core Details Corrective Action	As a form creator, I want to enter the Corrective Action so that similar issues can be prevented in the future.	- The system must allow the form creator to enter the corrective action. The system must rephrase the entered text into a clear, grammatically correct, and well-structured sentence using AI, even if only a short sentence is entered. - The form creator must have the option to Accept, Edit, or Reject the AI suggestion before final submission.
FR-11	Before/After Photographs	As a form creator, I want to upload Before and After photographs so that I can demonstrate how the corrective action was implemented.	- The system must allow the upload of Before photograph and After photograph. - The system must allow the upload of up to 5 photographs per category (Before/After) to capture multiple angles. - The system must accept only supported image formats.
FR-12	Before/After Photographs Remarks	As a form creator, I want to enter remarks after uploading the photographs so that I can provide additional context for the images.	- The system must allow the form creator to enter remarks for the uploaded photographs. - The remarks must be saved successfully.
FR-13	Submission & Approval	As a form creator, I want to submit the form with my Name, Designation, Signature, and Date so that it can be sent for approval.	- The system must capture the creator's Name, Designation, Signature, and Date. - The system must allow submission only after all mandatory fields are completed.
FR-14	Submission & Approval	As an approver, I want to review the submitted Lesson Learned form so that I can approve or reject it.	- The system must display a preview of all submitted information. - The approver must be able to Approve or Reject the form. - If rejected, the system must route the form back to the creator with comments.
FR-15	Escalation	As a Lesson Learned workflow, I want to trigger automated email alerts whenever the approval timeline is breached so that delays are escalated to top management.	- The system must automatically identify approval SLA breaches. - The system must send email notifications to the designated escalation recipients. - The notification must include the reference number, pending approver, delay duration, and a link to the application.
FR-16	Reports	As a user, I want to export the Lesson Learned form into standard layouts so that I can present formatted documents during client review	The system must provide an Export to PDF option.

			The generated PDF must match the approved template and formatting.
FR-17	Notifications	As a user, I want to receive push notices or automated emails whenever a workflow phase updates, so that I can take immediate action.	The system must immediately trigger a tailored push alert or email detailing the project code, required action, and direct application link to the designated user profile.
FR-18	Offline Capability	As a form creator, I want to submit the form even without internet connection so that the application works fine in remote sites.	- The mobile application must support offline data entry. - It shall cache form inputs locally, capture the timestamp and last-known GPS coordinates at the exact time of record creation, and automatically sync the submission to the server once internet connectivity is restored.

12. Reports

All reports must incorporate BU/Project and process-specific filters to enable clear data generation.

S.No	Report
1	Lesson Learned Log
2	Lesson Learned Form Report
3	Escalation Summary

13. Deliverables

#	Deliverable	Description
1	Web Application	Fully functional web portal with all modules as per this SOW
2	Mobile Application	Fully functional mobile portal with all modules as per this SOW
3	Report Generator	Report module replicating the Digital Team approved template. PDF and Excel export capability for reports and raw data
4	AI Suggestion	Auto rephrasing by AI models for the Root Cause, Correction & Corrective Action
5	Approval Workflow	Submission, approval, and send-back workflow with notification system.
6	Master Data & Admin Panel	Setup and configuration of all master data per Section 10.
7	Role-Based Access & User Setup	Roles, permissions, and user onboarding per Section 9.

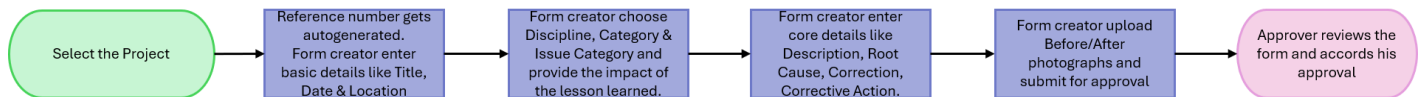
14. Assumptions

- Master data (Business Units, Projects, Users, etc.) will be provided by Digital team as and when new project comes.

- Report sample will be provided by the Digital Team; the development team will replicate the approved report.
- Mobile application will be developed for Android and iOS
- The development team and digital team will be available for requirement clarifications, reviews, and timely approvals throughout the project lifecycle.

15. Appendix – Understanding Document

15.1 Flowchart



16. Sign-Off & Approvals

This document requires formal review and approval from the Digital Team and Quality team before development and deployment.

Prepared By

Name	Department	Signature	Date
Subbu Sankar V R	Digital Team	V.R. Subbu Sankar	02-07-2026

Approved By (Quality Team)

Name	Department	Signature	Date
Mohammed Attaullah	Quality		
Fouad Saeed Al-Zahrani	Quality		
Owies Alrababah	Quality		